Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Applied Business Analytics

Task 3

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Task (on house_price_regression_dataset.csv) (Don't include in print document)

Q.1 Find the correlation between Year_Built and House_Price.

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
df = pd.read_csv("house_price_regression_dataset.csv")
print(df.head())
print(df.columns)
```

```
   Square_Footage  Num_Bedrooms  Num_Bathrooms  Year_Built  Lot_Size  \
0            1360             2              1        1981  0.599637
1            4272             3              3        2016  4.753014
2            3592             1              2        2016  3.634823
3             966             1              2        1977  2.730667
4            4926             2              1        1993  4.699073

   Garage_Size  Neighborhood_Quality   House_Price
0            0                     5  2.623829e+05
1            1                     6  9.852609e+05
2            0                     9  7.779774e+05
3            1                     8  2.296989e+05
4            0                     8  1.041741e+06
Index(['Square_Footage', 'Num_Bedrooms', 'Num_Bathrooms', 'Year_Built',
       'Lot_Size', 'Garage_Size', 'Neighborhood_Quality', 'House_Price'],
```

```
correlation = df["Year_Built"].corr(df["House_Price"])
print("Correlation between Year_Built and House_Price:")
print(correlation)
```

```
Correlation between Year_Built and House_Price:
0.051967361931064486
```

Q.2 Create a correlation matrix for all numerical columns.

```
correlation_matrix = df.corr(numeric_only=True)
print("Correlation Matrix:")
print(correlation_matrix)
```

```
Correlation Matrix:
                      Square_Footage  Num_Bedrooms  Num_Bathrooms  Year_Built  \
Square_Footage              1.000000     -0.043564      -0.031584   -0.022392
Num_Bedrooms               -0.043564      1.000000       0.022848   -0.015820
Num_Bathrooms              -0.031584      0.022848       1.000000   -0.021063
Year_Built                 -0.022392     -0.015820      -0.021063    1.000000
Lot_Size                    0.089479     -0.009355       0.034923   -0.061050
Garage_Size                 0.030593      0.113761       0.024846   -0.025485
Neighborhood_Quality       -0.008357     -0.049024       0.017585   -0.009549
House_Price                 0.991261      0.014633      -0.001862    0.051967

                      Lot_Size  Garage_Size  Neighborhood_Quality  House_Price
Square_Footage        0.089479     0.030593             -0.008357     0.991261
Num_Bedrooms         -0.009355     0.113761             -0.049024     0.014633
Num_Bathrooms         0.034923     0.024846              0.017585    -0.001862
Year_Built           -0.061050    -0.025485             -0.009549     0.051967
Lot_Size              1.000000     0.002436              0.037630     0.160412
Garage_Size           0.002436     1.000000             -0.011287     0.052133
Neighborhood_Quality  0.037630    -0.011287              1.000000    -0.007770
House_Price           0.160412     0.052133             -0.007770     1.000000
```

Q.3 Perform linear regression using:X = Year_Built Y = House_Price Display the coefficient and intercept.

```
X = df[["Year_Built"]]
Y = df["House_Price"]
model = LinearRegression()
model.fit(X, Y)
coefficient = model.coef_[0]
intercept = model.intercept_
print("Linear Regression: Year_Built -> House_Price")
print("Coefficient:", coefficient)
print("Intercept:", intercept)
print("\nRegression Equation:")
print(f"House_Price = {coefficient:.4f} * Year_Built + {intercept:.4f}")
```

```
Linear Regression: Year_Built -> House_Price
Coefficient: 638.6524884727304
Intercept: -649854.0823295026

Regression Equation:
House_Price = 638.6525 * Year_Built + -649854.0823
```

Q.4 Calculate the R2 score for the regression model: Year_Built - House_Price

```
X = df[["Year_Built"]]
Y = df["House_Price"]
model = LinearRegression()
model.fit(X, Y)
Y_pred = model.predict(X)
r2 = r2_score(Y, Y_pred)
print("R2 Score: Year_Built -> House_Price")
print(r2)
```

```
R2 Score: Year_Built -> House_Price
0.0027006067060743044
```

Q.5 Calculate the R2 score for:Square_Footage - House_Price

```
X = df[["Square_Footage"]]
Y = df["House_Price"]
model = LinearRegression()
model.fit(X, Y)
Y_pred = model.predict(X)
r2 = r2_score(Y, Y_pred)
print("R2 Score: Square_Footage -> House_Price")
print(r2)
```

```
R2 Score: Square_Footage -> House_Price
0.9825992634147099
```

Q.6 Calculate the R2 score for: Neighborhood_Quality - House_Price

```
X = df[["Neighborhood_Quality"]]
Y = df["House_Price"]
model = LinearRegression()
model.fit(X, Y)
Y_pred = model.predict(X)
r2 = r2_score(Y, Y_pred)
print("R2 Score: Neighborhood_Quality -> House_Price")
print(r2)
```

```
R2 Score: Neighborhood_Quality -> House_Price
6.037338916142776e-05
```

Q.7 Compare the R2 scores of all three regression models.

```python
# Model 1: Year_Built -> House_Price
X1 = df[["Year_Built"]]
Y1 = df["House_Price"]
model1 = LinearRegression()
model1.fit(X1, Y1)
r2_1 = r2_score(Y1, model1.predict(X1))
# Model 2: Square_Footage -> House_Price
X2 = df[["Square_Footage"]]
Y2 = df["House_Price"]
model2 = LinearRegression()
model2.fit(X2, Y2)
r2_2 = r2_score(Y2, model2.predict(X2))
# Model 3: Neighborhood_Quality -> House_Price
X3 = df[["Neighborhood_Quality"]]
Y3 = df["House_Price"]
model3 = LinearRegression()
model3.fit(X3, Y3)
r2_3 = r2_score(Y3, model3.predict(X3))
# Compare R2 Scores
r2_scores = pd.DataFrame({
    "Regression Model": [
        "Year_Built -> House_Price",
        "Square_Footage -> House_Price",
        "Neighborhood_Quality -> House_Price"
    ],
    "R2 Score": [
        r2_1,
        r2_2,
        r2_3
    ]
})
print("Comparison of R2 Scores:")
print(r2_scores)
print("\nR2 Scores from highest to lowest:")
```

```
Comparison of R2 Scores:
                      Regression Model  R2 Score
0            Year_Built -> House_Price  0.002701
1        Square_Footage -> House_Price  0.982599
2  Neighborhood_Quality -> House_Price  0.000060

R2 Scores from highest to lowest:
                      Regression Model  R2 Score
1        Square_Footage -> House_Price  0.982599
0            Year_Built -> House_Price  0.002701
2  Neighborhood_Quality -> House_Price  0.000060
```

Compare the R2 scores of all three regression models.

Regression Model	R^2 Score	Percentage Explained
Year_Built → House_Price	0.0027	0.27%
Square_Footage → House_Price	0.9826	98.26%
Neighborhood_Quality → House_Price	0.0000604	0.01%